Criticality and phase diagram study of the long-range quantum Ising chain

Korrelationstage 2021

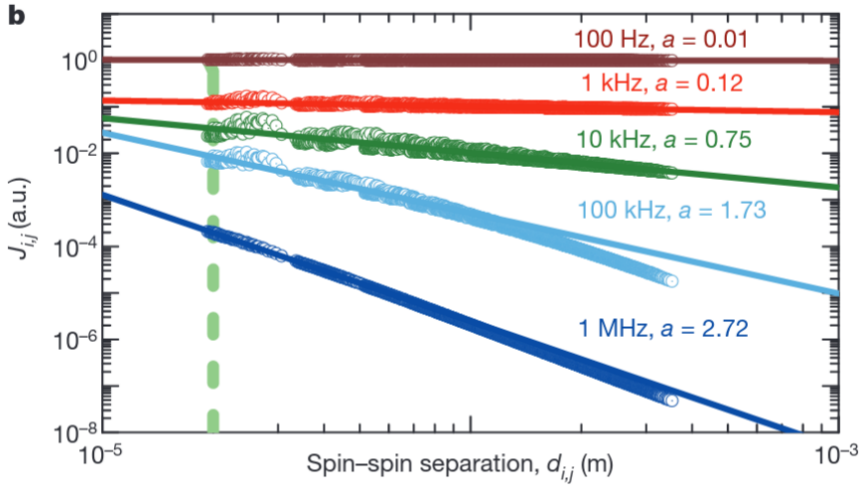
Eduardo Gonzalez Lazo

Introduction

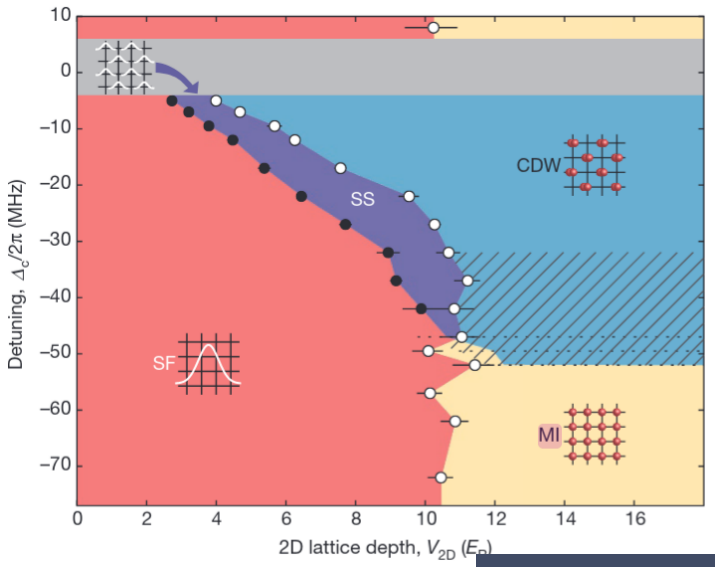
Long Range(LR) interacting quantum systems

- Power-law-decaying interactions can be engineered in experiments
- Analytically and numerically challenging to study
- Linked to interesting many-body phenomena (e.g., supersolidity)
- Many basic results in statistical mechanics cannot be applied (e.g., additivity, Mermin-Wagner theorem...)

Be ions stored in a Penning trap



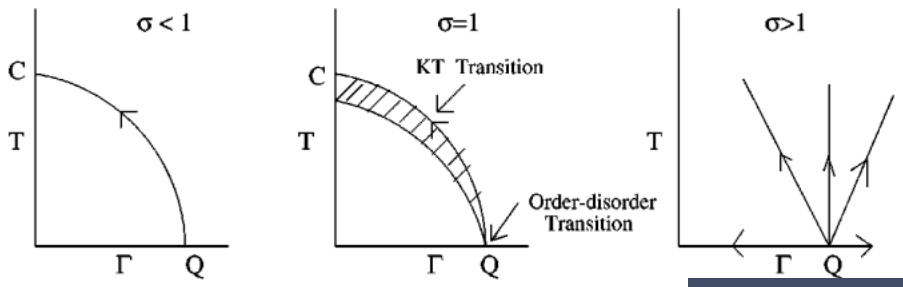
[Britton et al., 2012]



[Landig et al., 2016]

Introduction

$$\hat{H} = -\sum_{ij} J_{ij}\sigma_i^z\sigma_j^z - \Gamma\sum_i \sigma_i^x \qquad J_{ij} = \frac{J}{r_{ij}^{1+\sigma}},\ \alpha = 1+\sigma$$



[Dutta and Bhattacharjee, 2001]

QPT

- ☐ $\alpha \leq 5/3$
- ☐ $5/3 < \alpha < 3$
- ☐ $\alpha \geq 3$

CPT

- ☐ $\alpha \leq 3/2$ → Mean-Field
- ☐ $3/2 < \alpha \leq 2$ → Nontrivial
- ☐ $\alpha > 2$ → Short Range (SR)

We performed an in-depth study of the critical properties of the system, focusing on the finite temperature critical line

The model

System Hamiltonian

$$\hat{H} = -\frac{V}{K}\sum_{i<j}\frac{\hat{S}_i^z\hat{S}_j^z}{r_{ij}^{\alpha}} - h\sum_i \hat{S}_i^x.$$

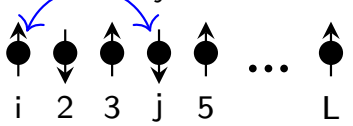
$$K = \sum_{i<j} r_{ij}^{-\alpha}$$

$$\hat{S}_i^x = \frac{\hbar}{2}\sigma_i^x$$

Reduced units Hamiltonian

$$\hat{H} = -\sum_{i<j}\frac{V}{K}\frac{\hat{S}_i^z\hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

$$J_{ij} = \frac{V}{r_{ij}^{\alpha}}$$



with PBC

Path Integral Monte Carlo

PIMC

Numerically exact technique for the study of unfrustrated systems of bosons and quantum spins.

- Maps to a classical, anisotropic, 2D Ising model (short range & long-range interactions)
- Properties of the quantum system are accessed via conventional Metropolis-type Monte Carlo

Our implementation

Combine:

- Wolff cluster updates [Wolff, 1989] (short-range part)
- Long-range cluster updates [Luijten and Blöte, 1997] (long-range part)

Criticality and finite-size systems

PIMC gives direct access to observables like

Binder cumulant

$$U = \frac{1}{2}\left[3 - \frac{\langle m_z^4 \rangle}{\langle m_z^2 \rangle^2}\right]$$

Susceptibility

$$\chi = \beta L(\langle m_z^2 \rangle - \langle |m_z| \rangle^2)$$

The finite-size scaling hypothesis

$$|t| \sim \xi^{-1/\nu}$$
$$Q \sim \xi^{\kappa/\nu}$$

$$|t_{max}(L)| \sim L^{-1/\nu}$$
$$Q_{max}(L) \sim L^{\kappa/\nu}$$

The hypothesis

$$Q(t, L) = L^{\sigma} f(\xi/L)$$

Ground state phase transition

Hamiltonian

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

Binder cumulant

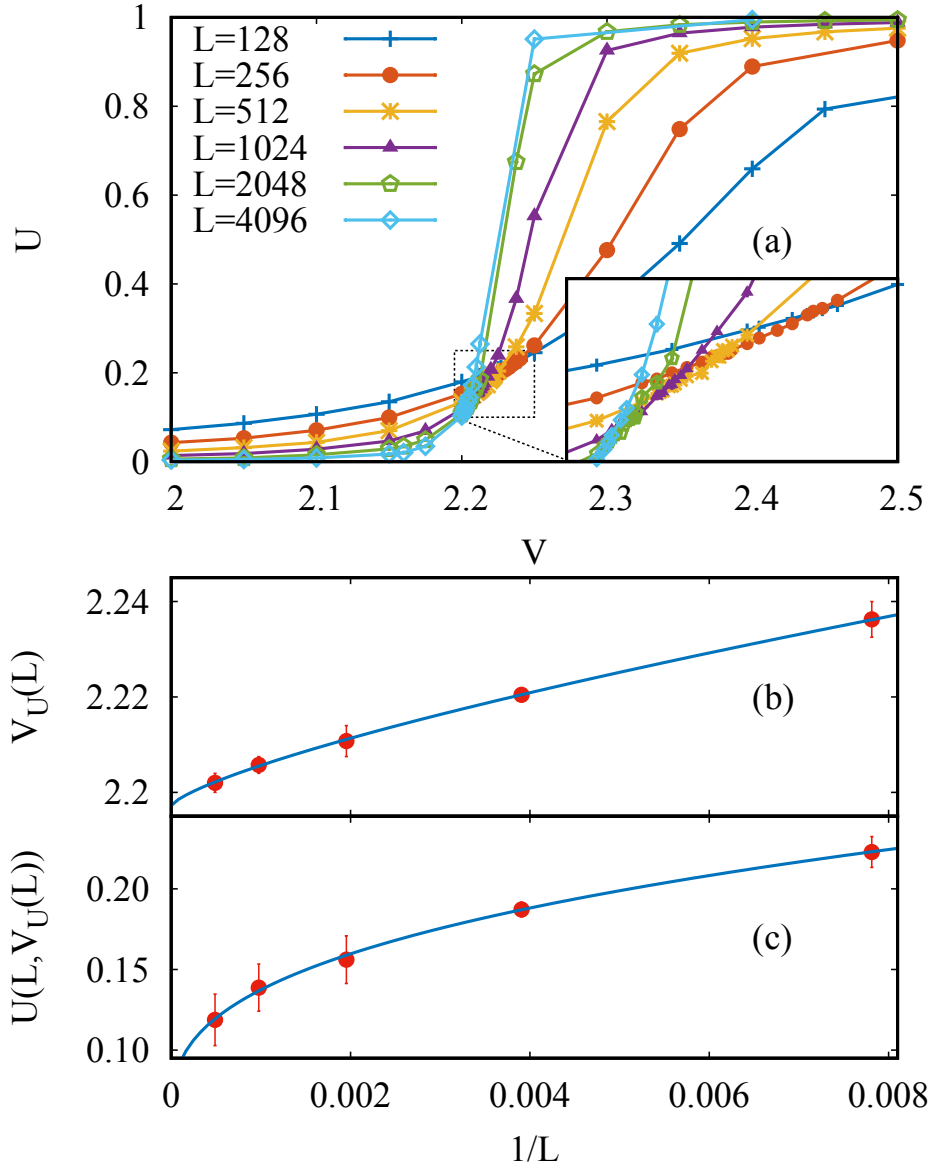
$$U = \frac{1}{2}\left[3 - \frac{\langle m_z^4 \rangle}{\langle m_z^2 \rangle^2}\right]$$

FSS

$$V_U(L) = V_c\left(1 + aL^{-\omega-\frac{1}{\nu}}\right)$$

$$U(L, V_U(L)) = b + cL^{-\omega}$$

[Angelini et al., 2014]



Finite temperature phase transition

Hamiltonian

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

Binder cumulant

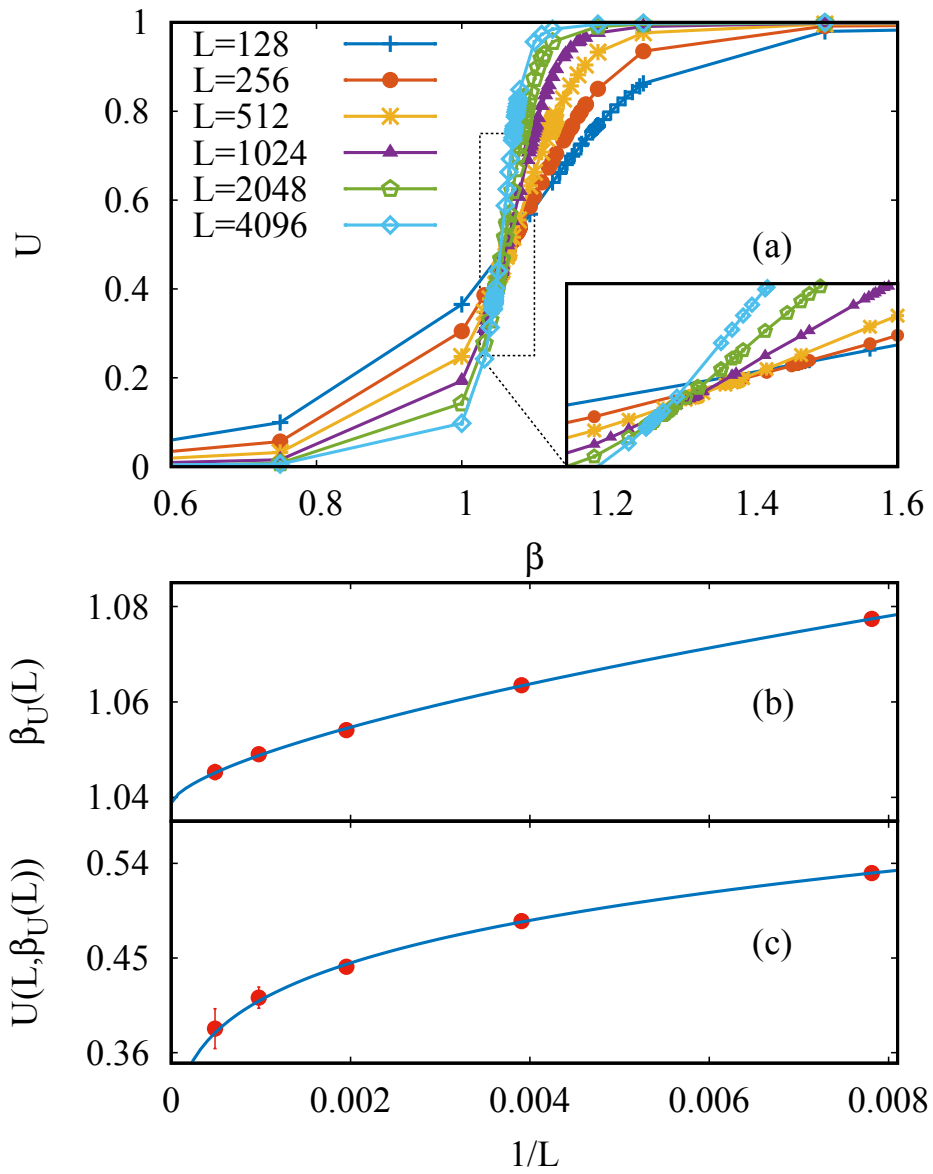
$$U = \frac{1}{2}\left[3 - \frac{\langle m_z^4 \rangle}{\langle m_z^2 \rangle^2}\right]$$

FSS

$$\beta_U(L) = \beta_c \left(1 + aL^{-\omega - \frac{1}{\nu}}\right)$$

$$U(L, \beta_U(L)) = b + cL^{-\omega}$$

[Angelini et al., 2014]



Finite temperature phase transition

Hamiltonian

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

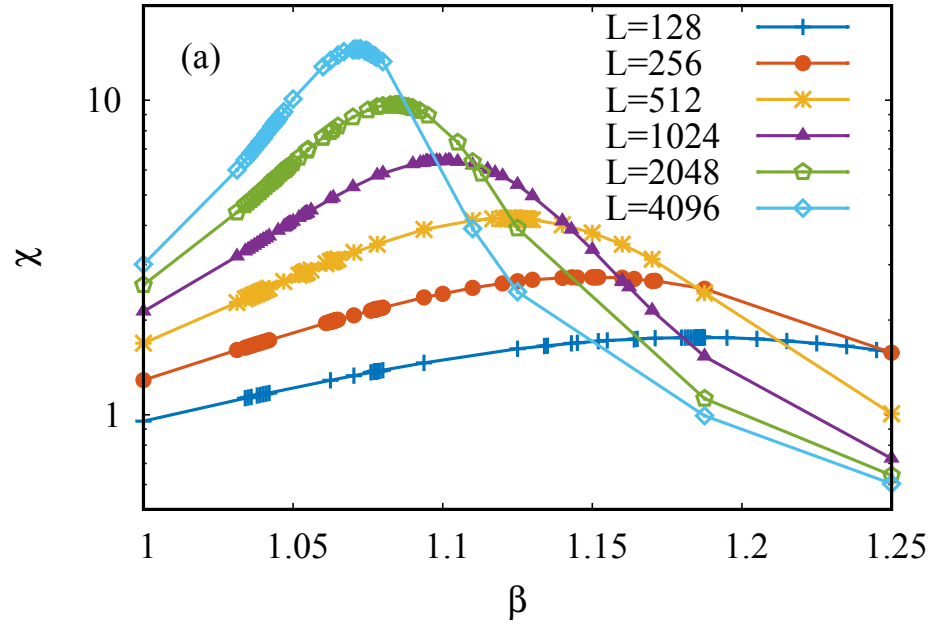
Susceptibility

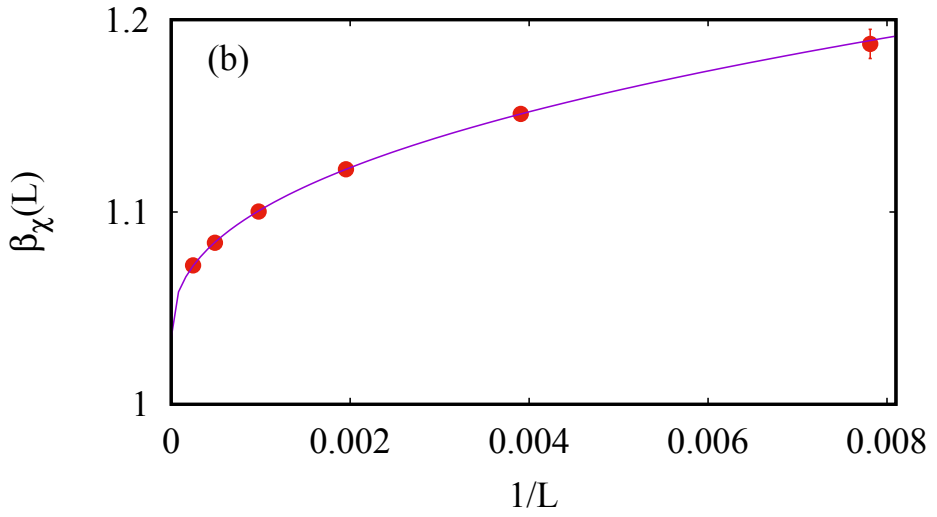
$$\chi = \beta L(\langle m_z^2 \rangle - \langle |m_z| \rangle^2)$$

FSS

$$\beta_\chi(L) = \beta_c + fL^{-1/\nu}$$

[Sandvik, 2010]





Finite temperature phase transition

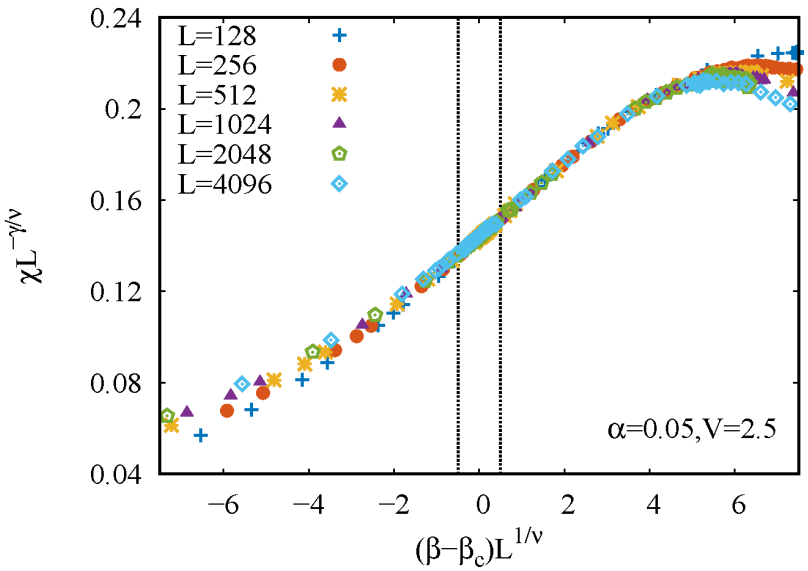
Curve collapse analysis

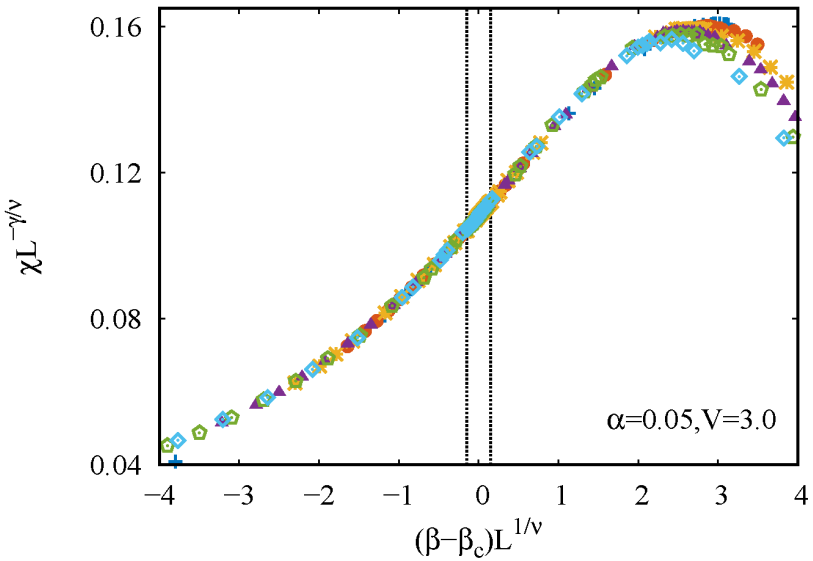
$$\chi \sim L^{\frac{\gamma}{\nu}} \cdot f(L^{-\frac{1}{\nu}}(\beta - \beta_c))$$

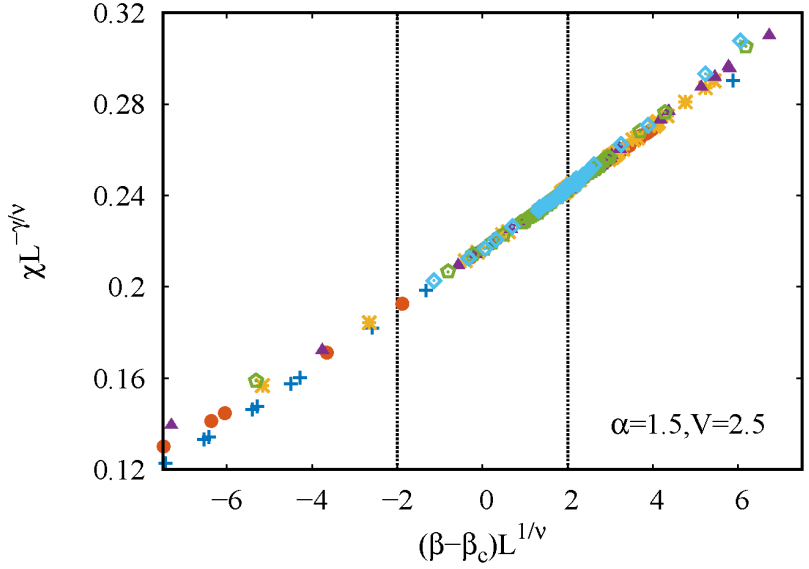
Perform a high order polynomial fit

$$y_l \equiv \chi(L) L^{-\gamma/\nu}$$
$$x_l \equiv (\beta - \beta_c) L^{1/\nu}$$

[Sandvik, 2010]







Criticality and phase diagram

Hamiltonian

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^\alpha} - \sum_i \hat{S}_i^x$$

Ground state results

α	V_c	$1/\nu$
0.05	1.9997(4)	0.50(7)
1.50	2.1972(7)	0.39(6)

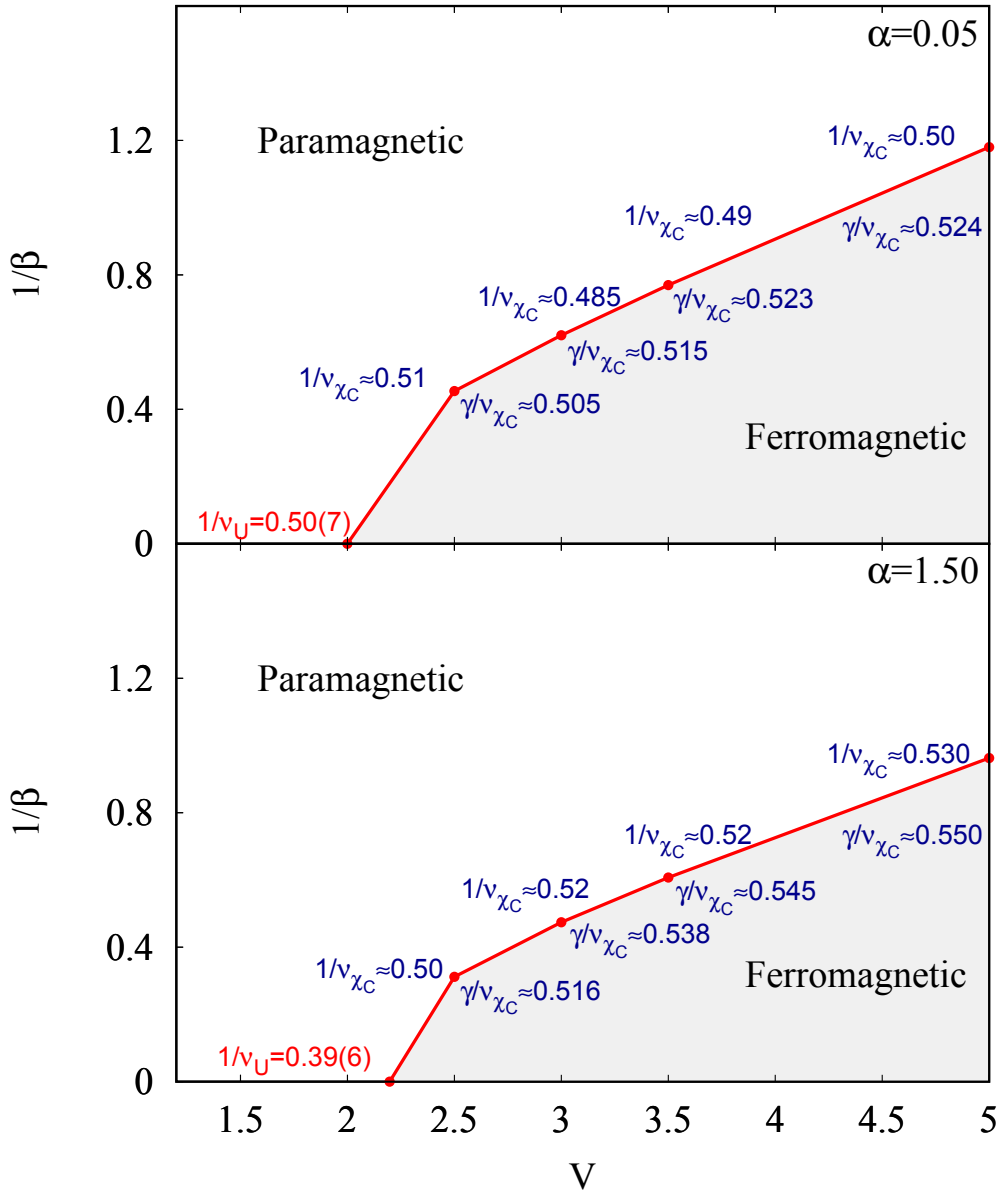
Finite Temperature results

		β_c			$1/\nu$			γ/ν
	V	U	χ	χ_{cc}	U	χ	χ_{cc}	χ_{cc}
$\alpha = 0.05$	$V = 2.5$	2.2007(4)	2.23(1)	2.2000	/	0.72(4)*	0.51	0.505
	$V = 3.0$	1.6120(7)	1.61(1)	1.612	/	0.54(3)	0.485	0.515
	$V = 3.5$	1.299(1)	1.303(3)	1.303	/	0.54(2)	0.49	0.523
	$V = 5.0$	0.8474(2)*	0.844(2)	0.8491	0.5(1)	0.47(2)	0.50	0.524
$\alpha = 1.50$	$V = 2.5$	3.21(1)	3.351(9)	3.229	0.49(7)	0.75(1)*	0.50	0.516
	$V = 3.0$	2.109(1)*	2.12(1)	2.115	0.50(2)	0.48(3)	0.52	0.538
	$V = 3.5$	1.647(6)	1.646(5)	1.650	0.5(2)	0.46(2)	0.52	0.545
	$V = 5.0$	1.039(1)	1.035(1)	1.041	0.44(7)	0.41(1)	0.530	0.550

Criticality and phase diagram

Hamiltonian

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$



Conclusions

- We study the quantum Ising chain with algebraically-decaying interactions
- We investigate the previously unexplored $\alpha < 1$ regime
- Preliminary results:
  - Finite-T transition at $\alpha = 0.05$ compatible with the ground-state universality class
  - Situation less clear for $\alpha = 1.50$, further study required

Acknowledgements

Adriano Angelone

Marcello Dalmonte

Angelo Russomanno

Markus Heyl

Thank you for your attention

Give feedback:

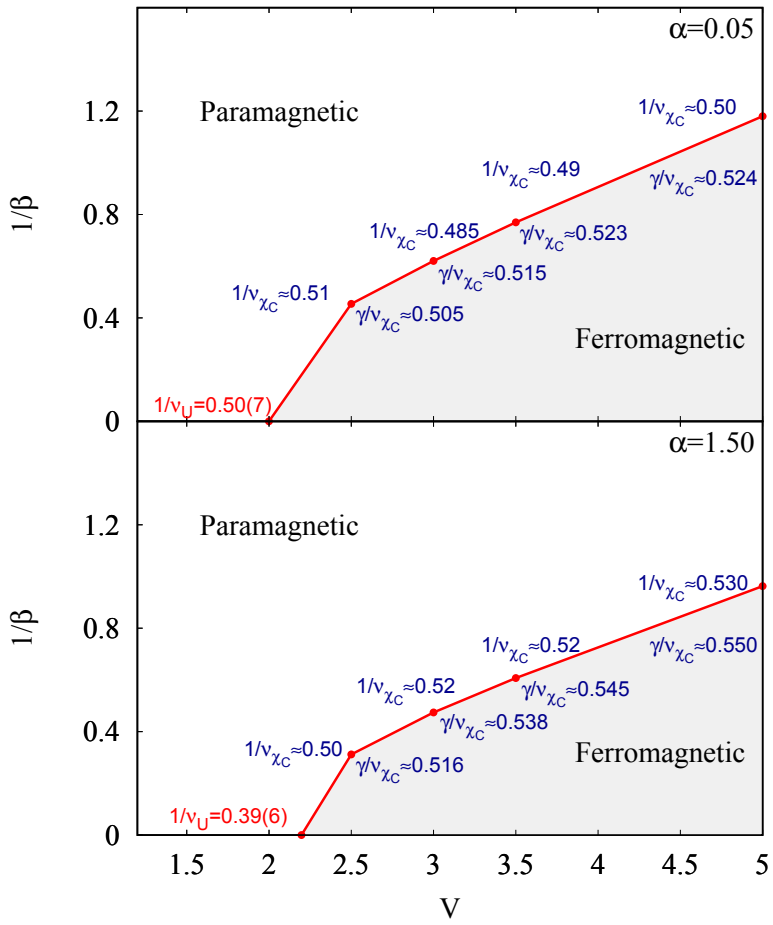
Download:

Bibliography

- Angelini, M. C., Parisi, G., and Ricci-Tersenghi, F. (2014). Relations between short-range and long-range Ising models. *Physical Review E*, 89(6):062120.
- Britton, J. W., Sawyer, B. C., and Keith, A. C. (2012). Engineered two-dimensional ising interactions in a trapped-ion quantum simulator with hundreds of spins. *Nature*, 484:489 – 492.
- Dutta, A. and Bhattacharjee, J. K. (2001). Phase transitions in the quantum ising and rotor models with a long-range interaction. *Phys. Rev. B*, 64:184106.
- Landig, R., Hruby, L., and Dogra, N. (2016). Quantum phases from competing short- and long-range interactions in an optical lattice. *Nature*, 532:476 – 479.
- Luijten, E. and Blöte, H. W. J. (1997). Classical critical behavior of spin models with long-range interactions. *Phys. Rev. B*, 56:8945–8958.
- Sandvik, A. W. (2010). Computational studies of quantum spin systems. *AIP Conference Proceedings*, 1297(1):135–338.
- Wolff, U. (1989). Collective monte carlo updating for spin systems. *Phys. Rev. Lett.*, 62:361–364.

Criticality and phase diagram study of the long-range quantum Ising chain

Eduardo Gonzalez Lazo

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

- Ising spin chains with long range interactions display interesting physics and are challenging to study
- We use Path Integral Monte Carlo methods:exact results at large sizes
- We investigate universal and nonuniversal features for several values of α